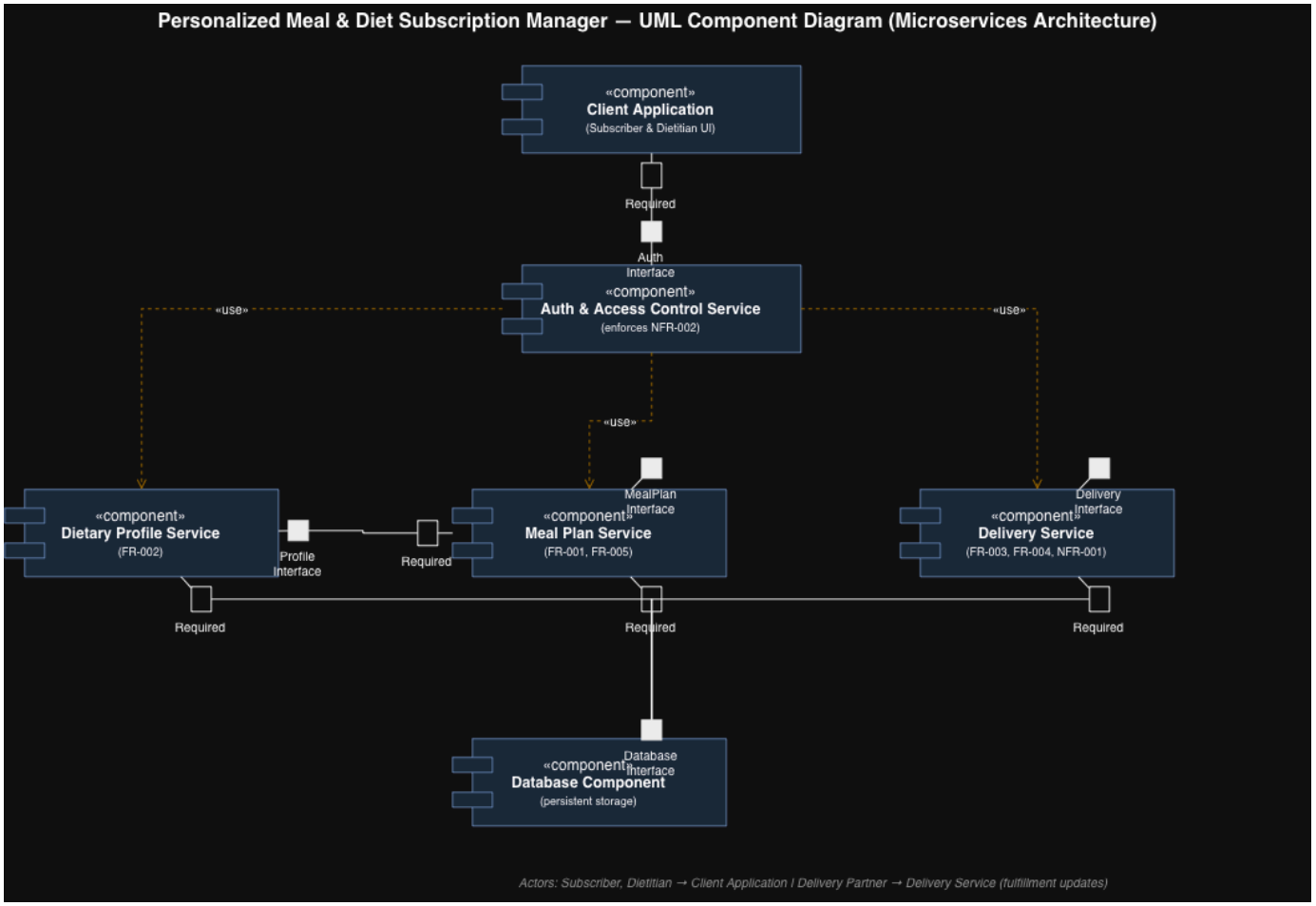
# SE LAB - 3

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Problem Statement - 19

## Personalized Meal & Diet Subscription Manager

## Component Diagram



# Justification:

### Architecture Selection

We chose **Microservices Architecture** for the Personalized Meal & Diet Subscription Manager System.

### Architectural Choice

The system is decomposed into independent services : Dietary Profile Service, Meal Plan Service, Delivery Service, and an Auth & Access Control Service  coordinated through an API gateway, with each service owning its own data access to a shared Database Component.

### Two Reasons

First, the system's core functions have distinct workload profiles: Meal Plan Service performs compute-intensive constraint validation and must regenerate plans when allergen or macro checks fail (per FR-001's alternate flow), while Delivery Service handles lighter, time-sensitive scheduling operations gated by a strict 12-hour cutoff (NFR-001). Decoupling these into independent services allows each to scale and evolve without impacting the other.

Second, the system serves three distinct actor types : Subscriber, Dietitian, and Delivery Partner, each with different access patterns and update frequencies. Microservices let each service expose only the interface relevant to its consumers, rather than forcing all actors through one monolithic interface.

### Security Advantage

A dedicated Auth & Access Control Service authenticates and authorizes every request before it reaches the Dietary Profile or Meal Plan services, directly enforcing NFR-002's requirement to restrict sensitive dietary and health data to authenticated, authorized users. Isolating this logic in its own service also limits the blast radius if any single service is compromised, since each microservice only exposes the data relevant to its function.

**Performance Benefit**

The Meal Plan Service, the most computationally demanding component due to constraint validation and regeneration can be scaled independently during peak load (e.g., weekly plan generation for all subscribers) without over-provisioning the lighter-weight Profile or Delivery services, improving resource efficiency across the system.